

CS 4343: DEEP LEARNING

Group Project Proposal Assignment

Fall A-Term 2026

Instructor: Professor Raha Moraffah

TEAM SIZE	DEADLINE	FINAL DELIVERABLE	LENGTH
4-5 students	Tuesday, Sept. 22, 2026	ONE PDF file per team	3-4 pages + references

Purpose. Your team will propose a focused deep-learning project that is technically meaningful, experimentally testable, and feasible within WPI A-Term. The proposal is a planning document: after reading it, the instructor should know exactly what problem you intend to solve, what model(s) you intend to train, what data you will use, how you will evaluate the work, and how the 4-5 team members will complete it. Final project materials are due Thursday, October 8, 2026, and team presentations are Friday, October 9, 2026.

1. What Counts as an Appropriate CS 4343 Project?

Your project must connect directly to course material and require substantive neural-network training and analysis. Appropriate technical connections include feedforward networks, convolutional neural networks (CNNs), recurrent networks (RNNs/LSTMs/GRUs), attention-based models, backpropagation and optimization, and training/regularization methods such as stochastic gradient descent, dropout, pooling, and batch normalization.

Required technical core: train or fine-tune at least one neural-network model and evaluate it quantitatively on a clearly defined task. The project should focus on the design, training, evaluation, and analysis of the proposed model. **LLMs and other generative-AI models may not be used as the project model, technical solution, or evaluation system.**

- State one or more testable research questions or hypotheses. Do not propose only “apply deep learning to dataset X.”
- Explain what quantitative evidence would indicate that the proposed model is successful or informative for the stated task.
- Define evaluation metrics before the final experiments, explain why those metrics match the task, and describe how you will analyze the resulting performance.
- Plan model-focused experiments or analyses that examine performance, training behavior, generalization, robustness, regularization, optimizer choices, ablations, or another justified aspect of the proposed model.
- Keep the scope realistic for the remaining A-Term and the compute available to your team.
- If you use a pre-trained model, your contribution must go beyond inference: fine-tune it, adapt it, or analyze specific architecture/training choices.

2. Proposal Deliverable: Use These Exact Sections

FINAL PROPOSAL DELIVERABLE: Submit exactly **ONE PDF file per team**. This single PDF must contain the complete proposal, including all required sections, references, and any required generative-AI disclosure appendix. Do not submit separate files. The main text should be approximately **3-4 pages**, excluding references. Figures and compact tables are encouraged when they clarify the model, data, evaluation, or timeline.

Section	Required content	Suggested space
Title + team	Descriptive project title; group/team name; full names of all 4-5 members; identify one group leader.	Header

Section	Required content	Suggested space
1. Problem and motivation	Define the problem precisely. Explain why it matters, why it is nontrivial, and why deep learning is appropriate. End with 1-3 research questions or hypotheses.	1/2 page
2. Proposed solution + course connection	Name the model family/architecture(s), inputs and outputs, loss function, optimizer/training strategy, and important regularization/design choices. Explicitly identify the CS 4343 concepts you will use or investigate.	3/4-1 page
3. Data plan	Give the dataset source, approximate size, features/labels, preprocessing, and train/validation/test plan. Confirm that the team can access the data. Note class imbalance, missing labels, licensing/privacy, or other data risks.	1/2 page
4. Model evaluation and analysis	Specify the primary evaluation metrics and explain how you will evaluate the proposed model. Define what quantitative evidence would count as a successful or informative result, and describe how you will analyze the results (for example, learning curves, class-wise performance, error/failure-case analysis, robustness, or ablations). Include expected tables/plots if known.	3/4 page
5. Feasibility, resources, and risks	Identify expected compute, software/frameworks, external dependencies, and the 1-2 biggest technical/data risks. Give a fallback plan that still produces a complete deep-learning project.	1/4 page
6. Team responsibilities	List every member and their initial technical responsibilities. Responsibilities may overlap, but each student must own substantive technical work; do not assign one member only writing/presentation duties.	Table/list
7. Project timeline	Provide a date-based or week-by-week plan through project completion. For each milestone, give an owner, expected output, and dependency. Reserve time for debugging, evaluation, error analysis, and writing.	1/2 page
References	Minimum 3 relevant sources: the dataset source, sources used to motivate the problem or model choice, and any important code/pre-trained model source used in the plan. Use one consistent citation style.	Not counted

Timeline requirement: your timeline must include at least five concrete milestones: data/preprocessing, model setup, first complete training run, model experiments/analysis, and final integration/evaluation. Every team member should appear as an owner or co-owner of at least one technical milestone.

Before You Submit: Proposal Checklist

- ☐ **Final submission is exactly ONE PDF file for the entire team**, due Tuesday, September 22, 2026. The PDF must contain the complete proposal, references, and any required generative-AI disclosure appendix; do not submit separate files.
- ☐ Project title; group/team name or number; full names of all 4-5 members; group leader.
- ☐ Clear problem, motivation, and 1-3 testable research questions/hypotheses.
- ☐ Specific deep-learning solution and explicit connection to CS 4343 material.
- ☐ Accessible dataset with preprocessing and train/validation/test plan.
- ☐ Quantitative metrics, clear evidence-of-success criteria, a model evaluation plan, and a plan for analyzing the results.
- ☐ Compute/resources, main risks, and a feasible fallback plan.
- ☐ Technical responsibilities for every team member.
- ☐ Concrete date-based/week-by-week timeline through the October 8 project deadline and October 9 presentation, with owners and expected outputs.
- ☐ At least three relevant references/sources, including the dataset source and any reused code or pre-trained model sources.
- ☐ If generative AI was used in preparing the proposal or project work, include the required disclosure: tool, exact prompts, where/how the output was used, and how it was validated, edited, and tested.
- ☐ Proposal proofread and written as a professional, coherent draft.

3. Curated Project Ideas Aligned with CS 4343

Here are some examples. Treat them as patterns to adapt, not assignments to copy.

Course area	Possible project	A reasonable core experiment
CNN / vision	Plant Disease Image Classification with CNNs	Train or fine-tune a CNN for the task; study augmentation, dropout, or batch normalization and analyze which choices improve validation/test performance.
CNN / vision	Recyclable Waste Classification under Lighting and Background Changes	Train a CNN and test robustness to background/lighting shift; study how augmentation or regularization affects the model under these shifts.
CNN / vision	Traffic Sign Recognition under Blur and Image Noise	Train a CNN with a justified regularization/augmentation strategy; report clean and corrupted-image performance and analyze failure modes.
CNN / vision	Static Hand-Sign Recognition	Train a compact CNN or fine-tune a pre-trained CNN; analyze class confusion and the effect of augmentation.
CNN / vision	Satellite Image Damage Classification after Natural Disasters	Use transfer learning for damage/no-damage or multi-class damage labels; study class weighting or augmentation choices and analyze class-wise performance.
CNN / vision	Small-Data Image Classification: Training from Scratch vs Transfer Learning	Choose a moderate image task, fine-tune a pre-trained CNN, vary training-set size, and analyze how model performance changes under limited data.
CNN / audio	Environmental Sound Classification from Spectrograms	Convert audio to spectrograms, train a 2D CNN, and analyze overall and per-class performance, confusion patterns, and sensitivity to preprocessing choices.
RNN / sequence	Human Activity Recognition from Wearable Sensors	Train an RNN/LSTM/GRU model on sensor windows; investigate sequence length or hidden-state size and analyze activity-wise performance.
RNN / sequence	Battery State-of-Health Prediction from Charge/Discharge Sequences	Train an LSTM or GRU sequence model; evaluate MAE/RMSE and analyze how prediction error changes across battery age.
RNN / sequence	Electricity Demand Forecasting with LSTM or GRU	Train an LSTM or GRU forecasting model; study look-back window and regularization and analyze forecast error across time periods.
RNN / sequence	Machine Fault Classification from Multivariate Sensor Sequences	Train a recurrent model for multivariate sensor sequences; analyze which fault classes are confused and how sequence length affects performance.
RNN + attention	Video Activity Classification from Precomputed Frame Features	Use fixed frame-level features to keep compute manageable; train an RNN/LSTM with temporal attention and analyze which frames contribute to predictions.
Text / attention	Review Sentiment Classification with LSTM/GRU or Attention	Use a manageable review dataset; train a recurrent or attention-based model and analyze errors by review length, sentiment strength, or ambiguity.
Text / attention	Topic Classification of News or Forum Posts	Train a recurrent or attention-based text classifier; study sequence length or regularization and analyze topic-wise errors.
Text / paired input	Duplicate Question Detection with a Shared Neural Encoder	Encode two questions with a shared neural encoder and predict whether they are duplicates; analyze similarity representations and difficult false-positive/false-negative cases.
Text / sequence	Online Review Helpfulness Prediction	Predict helpful/not-helpful or a helpfulness score using an RNN or attention-based model; analyze errors by review length, rating, or text characteristics.
Text / attention	Short-Text Intent Classification with Attention	Train an attention-based intent classifier; analyze errors on short, ambiguous, or rare intents and inspect attention behavior when appropriate.
Training study	What Regularization Helps Most? Dropout vs Batch Normalization vs Weight Decay	Hold the architecture/data fixed and compare regularization choices using learning curves, validation performance, and generalization gap.
Optimization study	SGD vs Adam: Optimizer and Learning-Rate Study	Hold model/data fixed; compare optimizers and learning-rate choices using convergence speed, stability, and final validation/test metrics.
Efficiency study	Small vs Large Neural Networks: Accuracy, Generalization, and Inference Cost	Train multiple model sizes on one task; compare accuracy/F1 plus parameter count, runtime, or memory.
Data study	Small-Data Generalization with Data Augmentation	Vary training-set size and augmentation/regularization; quantify when each technique improves generalization.

Some Example Projects That Need Revision Before Approval

- A software app, website, or user interface whose main contribution is engineering around a pre-existing model rather than deep-learning experimentation.
- A project with no held-out evaluation, no quantitative metric tied to the stated problem, or no plan to analyze whether the model addresses the research question.
- Training a very large model from scratch or proposing compute/data requirements that are unrealistic for the remaining A-Term.
- A project whose dataset is not yet accessible or requires most of the remaining term to collect/label.
- A project so broad that the team cannot name 1-3 concrete research questions and a feasible fallback plan.
- Use of private or sensitive personal data without a clear legal/ethical basis and instructor approval.

4. Recommended A-Term Project Timeline

Your proposal must contain your own timeline. The example below assumes the proposal is due September 22, the final project is due October 8, and team presentations are October 9. Adjust milestone ownership and pacing to your project, but reserve time for debugging, final evaluation, writing, and presentation preparation.

Dates	Milestone	Minimum expected output	Owner(s)
Sept. 22	Proposal	Approved problem, accessible data source, proposed deep model, metrics, success criteria, responsibilities, and repository/workspace.	All
Sept. 23-24	Data + model setup	Reproducible preprocessing/splits plus a working model/training pipeline; verify metric code and data leakage checks.	Named members
Sept. 28-30	First deep model	End-to-end training run with saved configuration, learning curves, and initial validation result.	Named members
Oct. 1-4	Experiments	Model-focused experiments or ablation(s); hyperparameter/regularization/robustness study; result and error analysis begins.	Split across team
Oct. 5-7	Final evaluation	Freeze model selection, evaluate once on test set, produce tables/figures, inspect failure cases.	All / analysis owners
Oct. 8	Final project due	Finalize experiments/results and submit the required final project materials. Ensure the repository/workspace and reported results are reproducible.	All
Oct. 9	Project presentation	Present the problem, proposed method, experiments, results, limitations, and main conclusions. All team members should be prepared to answer questions.	All

5. Generative AI Policy

LLMs and other generative-AI models may not be used as the project model, technical solution, evaluation system, or source of experimental results. If you use generative AI in preparing the proposal or in any other permitted supporting project work, you must document it clearly. Include the disclosure in an appendix; it does not count toward the 3-4 page proposal limit.

- State the tool you used.
- Include the exact prompts you entered.
- State exactly where and how you used the output.
- Explain how you validated, edited, and tested the result.

6. Proposal Grading Rubric - 100 Points

The proposal is graded on specificity, technical appropriateness, feasibility, and the quality of the execution plan. The criteria below describe what is required for **full credit** in each category. Scores within a category reflect how fully and clearly the proposal satisfies those criteria.

Criterion / points	Full Credit
Problem statement, motivation, and research question (15 points)	Clear, meaningful, appropriately scoped problem; strong motivation; explains why deep learning is appropriate; 1-3 specific research questions/hypotheses are testable.
Alignment with CS 4343 and technical depth (10 points)	Explicit connection to course concepts; project requires substantive neural-network training, experimentation, and analysis.
Proposed deep-learning solution (20 points)	Architecture/model family, inputs/outputs, loss, optimizer/training strategy, and key design/regularization choices are concrete and technically sensible.
Data plan (10 points)	Dataset access is confirmed; source, approximate size, labels/features, preprocessing, and train/validation/test strategy are clearly described; major data risks are addressed.
Model evaluation, evidence of success, and result analysis (15 points)	Appropriate quantitative metrics, a credible evaluation design, explicit success/informative-outcome criteria, and a concrete result-analysis plan directly test whether the proposed model addresses the research question.

Criterion / points	Full Credit
Timeline and team responsibilities (15 points)	Concrete milestones through completion; outputs, owners, dependencies, and time for debugging/evaluation/writing are included; all 4-5 members own substantive technical work.
Feasibility, resources, risks, and fallback plan (5 points)	Scope fits remaining A-Term and available compute; main risks are identified; fallback preserves a complete deep-learning project.
Organization, clarity, and professional presentation (5 points)	Coherent, concise, proofread, and easy to follow; tables/figures are labeled when used; terminology and formatting are consistent.
References and reproducibility planning (5 points)	Relevant sources are cited; the dataset and any reused code/pre-trained models are acknowledged; software/framework assumptions and a reproducible workflow are clear. If generative AI was used, the tool, exact prompts, where/how the output was used, and how the result was validated, edited, and tested are fully documented.
TOTAL: 100 points	

Approval gate. Regardless of the numerical score, a proposal may be returned for revision if it does not include substantive model training/fine-tuning, an accessible dataset, a meaningful evaluation and result-analysis plan, or a scope that can plausibly be completed by the end of A-Term.